

Introduction to Microeconomics

Lecture 13 & 14—Income and Substitution Effects, Expenditure Minimization, and ~~Price Indexes~~

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Income and Substitution Effects

A fall in price of a good has two effects:

- ① Consumers will tend to buy more of the good that has become cheaper and less of those goods that are now relatively more expensive. This response to a change in the relative prices of goods is called the **substitution effect**.
- ② Because one of the goods is now cheaper, consumers enjoy an increase in real purchasing power. The change in demand resulting from this change in real purchasing power is called the **income effect**.

Examples: an almost pure income effect and an almost pure substitution effect

- Suppose we are considering the rent that needs to be paid for living in an apartment versus other goods.
 - Suppose there's a decrease in the rental price of your unit.
 - Rent involves a **large share of your income**.
 - You will end up feeling “richer” because you will have more disposable income to buy other stuff.
 - This is an example of an (almost pure) income effect.
- What apple to buy? Granny Smith apples (green skin, crisp and juicy flesh, tart and acidic flavor) or McIntosh apples (red/green skin, tender white skin, tart flavor).
 - This purchase involves a very little share of your income.
 - Suppose that you just want an apple and thus, you are equally happy between both varieties.
 - You will end up buying the one that is cheaper.
 - If prices change, you buy the one with the lowest price, that is, you **substitute towards the relatively cheaper alternative**.
 - This is an example of an (almost pure) substitution effect.

Definitions

- The **substitution effect** is the change in consumption of a good associated with a change in its price, with the level of utility held constant.
- The **income effect** is the change in consumption of a good resulting from an increase in purchasing power, with relative prices held constant.

The Total effect of a price change and its decomposition

The total effect after a price change

- Suppose that we want to analyze the change in quantity demanded of good 1 after a price change from p_1^0 to p_1^1 , while holding the price of good 2 and income fixed. Then the **Total effect of the price change**, or TE , is

$$TE = x_1^m(p_1^1, \bar{p}_2, \bar{I}) - x_1^m(p_1^0, \bar{p}_2, \bar{I})$$

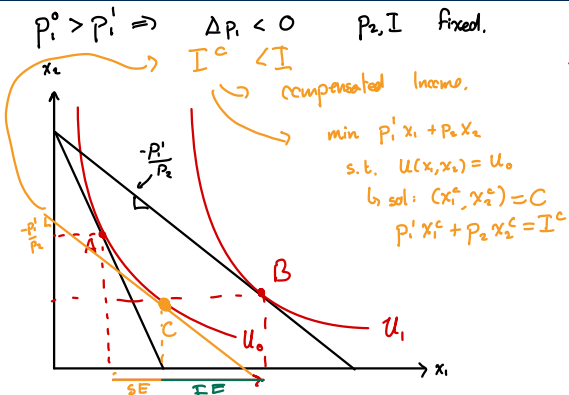
or for simplicity:

$$\Delta p_1 = p_1^1 - p_1^0$$

$$TE = x_1(p_1^1) - x_1(p_1^0) = SE + IE$$

- The goal is to decompose the TE into the substitution effect SE and the income effect IE .
 - The SE will respect the law of demand (goes in opposite direction to the price change).
 - The IE depends on whether the good is normal or inferior.

Graphical Analysis 1A: Normal good and a price decrease



$$A = (x_1^m(p_1^0), x_2^m(p_1^0))$$

$$B = (x_1^m(p_1^1), x_2^m(p_1^1))$$

$$C = (x_1^m(p_1^1, p_2, I^c), x_2^m(p_1^1, p_2, I^c))$$

$$TE = x_1^m(p_1^1) - x_1^m(p_1^0) > 0$$

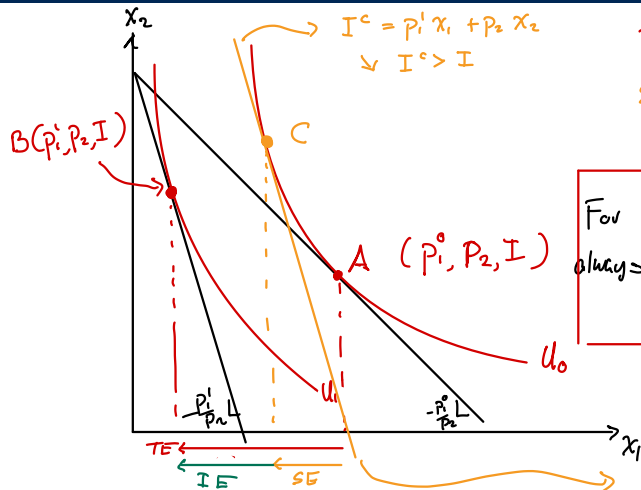
$$\begin{aligned}
 SE &= C - A \text{ (horizontal } x_1 \text{ coordinate)} \\
 &= x_1^m(p_1^1, p_2, I^c) - x_1^m(p_1^0, p_2, I) > 0
 \end{aligned}$$

$$\begin{aligned}
 IE &= B - C \\
 &= x_1^m(p_1^1, p_2, I) - x_1^m(p_1^1, p_2, I^c) > 0
 \end{aligned}$$

For a normal good, the TE will always satisfy the law of demand.

SE & IE have the same sign.

Graphical Analysis 1B: Normal good and a price increase



$$TE = B - A \quad (\text{horizontal movement}) < 0$$

$$SE = C - A < 0$$

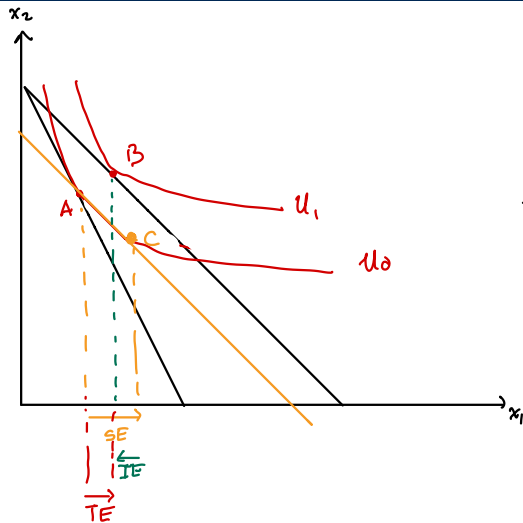
$$IE = B - C < 0$$

For a normal good, the TE will always satisfy the law of demand.
SE & IE have the same sign.

cost of C at new prices (p_1', p_2)

$$\begin{aligned} \min \quad & p_1' x_1 + p_2 x_2 \\ \text{s.t.} \quad & U(x_1, x_2) = U_0 \end{aligned} \quad \begin{aligned} I^c &= p_1' x_1^c + p_2 x_2^c \\ \hookrightarrow \text{sol. } C &= (x_1^c, x_2^c) \end{aligned}$$

Graphical Analysis 2A: Inferior good and a price decrease



$$SE = C - A > 0$$

$$IE = B - C < 0$$

$$TE = B - A > 0$$

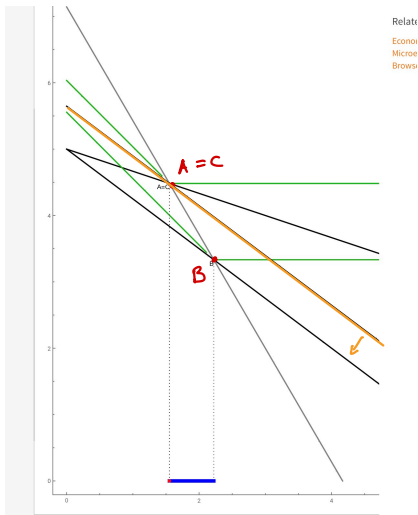
$$|SE| > |IE| \Rightarrow TE > 0$$

↳ SE dominates the income effect

⇒ TE satisfies the law of demand.

Graphical Analysis 2B: Inferior good and a price increase (homework!)

A theoretical Paradox: Giffen Goods and upward sloping demand.


$$\mathbb{I}^c > \mathbb{I}$$

Giffen paradox can only happen for inferior goods

$|SE| < |IE|$

$\Rightarrow$ TE violates
the law of demand.

The expenditure minimization problem

In order to be able to calculate analytically the decomposition of the total effect into the income and substitution effects, we will need to analyze “dual problem” to the utility maximization problem of the consumer. Recall:

$$\max_{x_1, x_2} u(x_1, x_2) \quad \text{subject to } p_1 x_1 + p_2 x_2 \leq I \quad (\text{UMP})$$

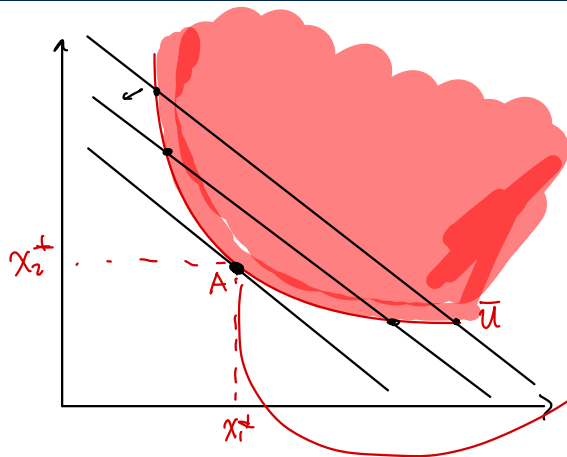
Expenditure Minimization

Given preferences, prices and a fixed level of utility $\bar{U}$, find the cheapest consumption bundle that achieves a level of utility of at least $\bar{U}$.

$$\min_{x_1, x_2} p_1 x_1 + p_2 x_2 \quad \text{subject to } u(x_1, x_2) \geq \bar{U} \quad (\text{EMP})$$

The parameters of this problem are p_1, p_2 and the utility level $\bar{U}$. The solutions to this problem, when prices change, will give an optimal bundle that achieves exactly the original utility level, but with a different price ratio.

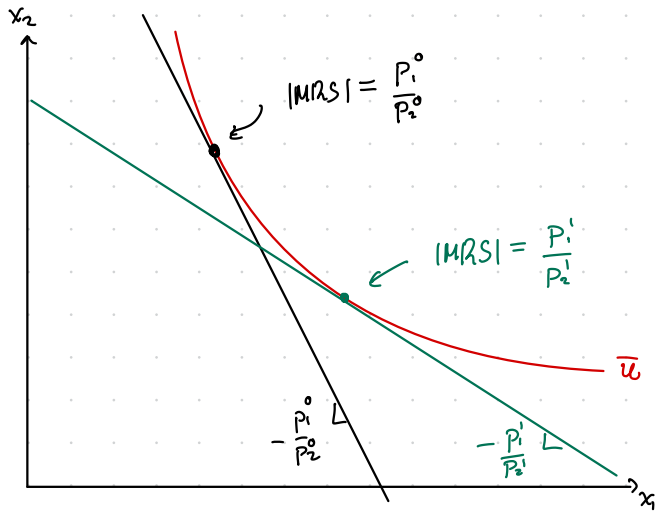
Graphical Analysis of the EMP and its optimality condition



Expenditures : $P_1 X_1 + P_2 X_2 = E$
iso-expenditures

$A = (x_1^+, x_2^+)$
achieves the minimal
expenditure:
 $E^+ = P_1 x_1^+ + P_2 x_2^+$
while giving utility
 $u(x_1^+, x_2^+) = \bar{u}$

Optimality Condition : $MRS = \frac{P_1}{P_2}$



Steps to decompose the total effect into the income and substitution effects

Suppose that we hold income fixed at I and the price of good 2 at p_2 . Let the initial price of good 1 be p_1^0 which then changes to p_1^1 .

- 1 Compute the demand functions $x_1(p_1^0, p_2, I)$ and $x_2(p_1^0, p_2, I)$.
- 2 Compute the achieved level of utility at those demands: $\bar{U} = u(x_1(p_1^0, p_2, I), x_2(p_1^0, p_2, I))$.
- 3 Compute the new demands when the price of good one changes to p_1^1 : $x_1(p_1^1, p_2, I)$ and $x_2(p_1^1, p_2, I)$.
- 4 The total effect is $TE = x_1(p_1^1, p_2, I) - x_1(p_1^0, p_2, I)$
- 5 To find the income required to leave the consumer at the utility level $\bar{U}$ at the new prices, solve the expenditure minimization problem:

$$\min_{x_1, x_2} p_1^1 x_1 + p_2 x_2 \quad s.t. \quad u(x_1, x_2) = \bar{U}$$

- 6 The solution to the problem above are called compensated demands $x_1^c(p_1^1, p_2, \bar{U})$ and $x_2^c(p_1^1, p_2, \bar{U})$.
- 7 Compute the compensated level of income $I^c = p_1^1 x_1^c + p_2 x_2^c$.
- 8 Note that $x_1^m(p_1^1, p_2, I^c) = x_1^c(p_1^1, p_2, \bar{U})$.
- 9 The substitution effect is $SE = x_1^c(p_1^1, p_2, \bar{U}) - x_1^m(p_1^0, p_2, I)$.
- 10 The income effect is $IE = x_1^m(p_1^1, p_2, I) - x_1^c(p_1^1, p_2, \bar{U})$.

Example with Cobb-Douglas Utility

cost-of-living index or price index Ratio of the present cost of a typical bundle of consumer goods and services compared with the cost during a base period.

ideal cost-of-living price index Cost of attaining a given level of utility at current prices relative to the cost of attaining the same utility at base-year prices.

- The ideal cost-of-living price index takes into consideration that consumers **substitute** towards relatively cheaper consumption bundles of goods and services when facing different price ratios across periods.

Laspeyres and Paasche Indexes

For a set of goods and services $\mathcal{C} = \{c_1, c_2, \dots, c_k\}$, let

- $p_{c,t}$: prevailing price of good c in period t .
- $q_{c,t}$: quantity of c consumed in period t .

Let t_n denote the current time, and t_0 denote the **base year**.

- Laspeyres Index

$$\mathcal{L}(t_0, t_n, q_{c,t_0}) = \frac{p_{c_1,t_n} q_{c_1,t_0} + p_{c_2,t_n} q_{c_2,t_0} + \dots + p_{c_k,t_n} q_{c_k,t_0}}{p_{c_1,t_0} q_{c_1,t_0} + p_{c_2,t_0} q_{c_2,t_0} + \dots + p_{c_k,t_0} q_{c_k,t_0}}$$

- Paasche Index

$$\mathcal{P}(t_0, t_n, q_{c,t_n}) = \frac{p_{c_1,t_n} q_{c_1,t_n} + p_{c_2,t_n} q_{c_2,t_n} + \dots + p_{c_k,t_n} q_{c_k,t_n}}{p_{c_1,t_0} q_{c_1,t_n} + p_{c_2,t_0} q_{c_2,t_n} + \dots + p_{c_k,t_0} q_{c_k,t_n}}$$

- In general, a price-index depends on the base year t_0 , the current year t_n , and the basket of goods q_c .

The ideal cost of living index: example

	200	2010
Price of books	\$20.00	\$107.5
Number of books		
Price of good	\$2.00	\$3.00
Kg of food		
Expenditure		

The ideal cost of living index: using the expenditure minimization problem to find the bundle in the new year

	200	2010
Price of books	\$20.00	\$107.5
Number of books		
Price of good	\$2.00	\$3.00
Kg of food		
Expenditure		

The ideal cost of living index: Computing the index

	200	2010
Price of books	\$20.00	\$107.5
Number of books		
Price of good	\$2.00	\$3.00
Kg of food		
Expenditure		

Computing the Laspeyres Index

	200	2010
Price of books	\$20.00	\$107.5
Number of books		
Price of good	\$2.00	\$3.00
Kg of food		
Expenditure		

Computing the Paasche Index

	200	2010
Price of books	\$20.00	\$107.5
Number of books		
Price of good	\$2.00	\$3.00
Kg of food		
Expenditure		

- We have that in general:

$$\mathcal{P} \leq \mathcal{I} \leq \mathcal{L}$$

- Potential remedy is to compute **chain-weighted indexes**:
 - For the Laspeyres index:

$$\mathcal{L}(t_0, t_1, q_{c,t_0}) \times \mathcal{L}(t_1, t_2, q_{c,t_1}) \times \cdots \times \mathcal{L}(t_{n-1}, t_n, q_{c,t_{n-1}})$$

- For the Paasche index:

$$\mathcal{P}(t_0, t_1, q_{c,t_1}) \times \mathcal{P}(t_1, t_2, q_{c,t_2}) \times \cdots \times \mathcal{P}(t_{n-1}, t_n, q_{c,t_n})$$

- An example of the Laspeyres index is the CPI (Consumer Price index): Fix a base year basket and a base year and track its index across time. The CPI is used to measure inflation in all countries around the world.
- Example of a Paasche index: GDP Deflator, which is the ratio of Nominal GDP/Real GDP. As you can see this index fixes a base year and a current year basket.
 - Nominal GDP: value of goods and services at current prices.
 - Real GDP: value of goods and services at base year prices.